

MTH 421 practice problems for final exam

IMPORTANT NOTE: These problems only cover the material introduced *after* the second midterm exam. The final exam is cumulative, and approximately half of it will cover older material (including groups,) so you should also review the midterm exams and practice midterm exams.

Problem 1

Let U be a set and $V \subseteq U$. Define $f : \mathcal{P}(U) \rightarrow \mathcal{P}(V)$ by $f(A) = A \cap V$.

- (a) Show that f is a surjective homomorphism (of rings.) You may use Venn diagrams to argue that it is a homomorphism.
- (b) What is its kernel?
- (c) What can you conclude using the FHT?

Problem 2

Give an example of each of the following.

- (a) A non-commutative ring.
- (b) A commutative ring without unity.
- (c) A commutative ring with unity that is not an integral domain.
- (d) An integral domain that is not a field.
- (e) A field.

Problem 3

- (a) If A and B are commutative rings with unity, is $A \times B$ a commutative ring with unity?
- (b) If A and B are integral domains, is $A \times B$ an integral domain?

Explain negative answers only.

Problem 4

Prove that the set $A = \{a + 2^{1/3}b + 2^{2/3}c \mid a, b, c \in \mathbb{R}\}$ is a subring of $\mathbb{R}$.

Problem 5

- (a) Give an example of a nonzero nilpotent element of $\mathbb{Z}_{81}$.
- (b) Give an example of an idempotent element of $\mathbb{Z}_{21}$ besides 0 and 1.
- (c) What element of $\mathbb{Z}_3 \times \mathbb{Z}_7$ does your answer to part (b) correspond to under an isomorphism?

Briefly explain your answers.

Problem 6

Prove, using only the ring axioms, that for any ring element a , $0a = 0$, where 0 is the zero element.

Problem 7

Prove that if F is a field and A is a subring of F containing the unity element 1 of F , then A is an integral domain. Then give an example of such F and A where A is not a field.

Problem 8

Prove that if a and b are elements of an integral domain such that $a^2 = b^2$, then $a = b$ or $a = -b$.

Problem 9

Let A be a commutative ring and let J be the set of all nilpotent elements of A . Prove that J is an ideal of A . Hint: use the binomial theorem to expand $(a + b)^n$.

Problem 10

Let A be a ring and let J be a prime ideal of A . Prove that J is semiprime, meaning that for all $a \in A$ and $n \in \mathbb{Z}^+$, if $a^n \in J$, then $a \in J$. Hint: use induction.

Problem 11

Suppose that A is a nontrivial commutative ring with unity 1 and that J is a prime ideal of A . Prove that the quotient ring A/J is an integral domain.

Problem 12

Let A be a nontrivial Boolean ring with unity 1 (recall that a Boolean ring is one in which every element is idempotent.) Prove that A has characteristic 2.

Problem 13

Let A be a ring and let J be an ideal of A . Suppose that for all $a, b \in A$, $ab - ba \in J$. Prove that the quotient ring A/J is commutative.

Problem 14

Let $f : A \rightarrow B$ be a homomorphism of rings and let J be its kernel. Prove that J is an ideal of A .